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DETAIL SPECIFICATION

CABINETS, MONITOR, CY-7359/FSS-9(V)

CY-7360/FSS-9(V), AND CY-7361/FSS-9(V)

Inactive for new design after 4 August 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of monitor cabinets with integral power supplies and mounting facilities with interface connections for the status and alarm monitor modules. The cabinets are components of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

1.2 Classification. Monitor cabinets are of the following types, as specified (see 6.2):

Type A - One Zone Monitor Cabinet, CY-7359/FSS-9(V).
Type B - Five Zone Monitor Cabinet, CY-7360/FSS-9(V).
Type C - Twenty-five Zone Monitor Cabinet, CY-7361/FSS-9(V).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army CECOM, ATTN: AMSEL-LC-IEW-D-ED, 10115 Gridley Rd, Suite 228, Ft Belvoir, VA 22060-5849 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Government documents.

2.2.1 Standards and handbooks. The following standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-889	-	Dissimilar Metals.

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454	-	General Guidelines for Electronic Equipment.
MIL-HDBK-781	-	Reliability Test Methods, Plans and Environments for Engineering Development, Qualification and Production.

(Unless otherwise indicated, copies of the above standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13220E3900	-	Cabinet, Monitor, 1 Zone, CY-7359/FSS-9(V).
TA13220E3901	-	Power Supply, 1 Zone Monitor.
TA13220E3902	-	Signal Module Assembly.
TA13220E4000	-	Cabinet, Monitor, 5 Zone, CY-7360/FSS-9(V).
TA13220E4001	-	Power Supply, 5 Zone Monitor.
TA13220E4100	-	Cabinet, Monitor, 25 Zone, CY-7361/FSS-9(V).
TA13220E4101	-	Power Supply, 25 Zone Monitor.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: US Army CECOM, IEWS Directorate, ATTN: AMSEL-LC-IEW-D-CE, 10115 Gridley Rd, Suite 228, Ft Belvoir, VA 22060-5849.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ASQC Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes.
ISO 10012-1	-	Quality Assurance Requirements for Measuring Equipment. - Part 1: Metrological confirmation system for measuring equipment.

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(Application for copies should be addressed to the: American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036.)

AMERICAN WELDING SOCIETY (AWS)

D9.1 - Welding of Sheet Metal.

(Application for copies should be addressed to: American Welding Society, 550 NW LeJeune Rd, PO Box 351040, Miami, FL 33135.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The 1 zone, 5 zone, and 25 zone monitor cabinets shall be in accordance with TA13220E3900, TA13220E4000, and TA13220E4100, respectively, and as specified herein.

3.1.1 Major components. The major components of the monitor cabinets shall be as follows:

a. Signal module assembly: TA13220E3902.
b. Power supply: TA13220E3901, TA13220E4001, and TA13220E4101 for the 1, 5, and 25 zone monitor cabinets respectively.

When specified (see 6.2), any of the major components listed above shall be furnished separately.

3.1.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly and operation of the monitor cabinet. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The monitor cabinets shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the monitor cabinet may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically

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advantageous life cycle costs. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the monitor cabinets.

3.4 Environmental conditions. When tested as specified in 4.4.2.7, the monitor cabinets and major components, when furnished separately, shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.3.2), performance degradation, or malfunction.

3.4.1 Operational.

a. Temperature range of -20 to +150 °F.
b. Humidity up to 95 percent relative.

3.4.2 Non-operational condition and storage.

a. Temperature range of -30 to +165 °F.
b. Humidity up to 95 percent relative.
c. Shock: bench handling (major components only), equipment not operating.

3.5 Detailed technical requirements. The monitor cabinets and major components when furnished separately shall perform as specified herein under any of the operating test conditions specified herein.

3.5.1 Wiring. Monitor cabinet internal wiring and wiring harnesses shall be in accordance with the wire list on the applicable top assembly drawing (TA13220E3900, TA13220E4000, TA13220E4100). Transmission line brackets containing lightning and EMI suppression components shall be in accordance with applicable top assembly drawings.

3.5.2 Signal module assembly. The signal module assembly shall contain an audible signaling device and the monitor cabinet alternating current (ac) power status indicator lamps.

3.5.2.1 Audible signaling device. When actuated, the signaling device shall sound at a frequency of 2900 ±500 Hertz (Hz) and an amplitude not less than 60 decibels (db) referenced to 0.0002 microbars sound pressure level at a distance of 10 feet on-axis, using "A" frequency weighting. The signal device shall sound whenever less than 1 volt direct current (Vdc) is being applied to terminal board 1, terminal 1 (TB1-1), referenced to ground, of the signal module. It shall also sound whenever the ac power status indicator lamps are flashing (see 3.5.2.2). The audible signaling device shall be silent when both of the following conditions exist: a voltage of 4 to 5 Vdc is applied to TB1-1; the ac power status indicator lamps are continuously on or off (not flashing).

3.5.2.2 Monitor cabinet ac power status indicator lamps. When the monitor cabinet is operating on external 117 (+8, -7) volts ac (Vac) power, the ac power status indicator lamps shall be continuously lit. When the monitor unit is operating on standby power, the ac power status indicator lamps shall be continuously off. When the external ac power to the monitor cabinet is removed for a period of 3 ±0.5 seconds or when ac power is re-applied to the monitor cabinet after having been removed, the lamps shall begin to flash at a frequency of 4 ±0.5 Hz. When the "LAMP TEST/ACKNOWLEDGE" switch is momentarily set to "ACKNOWLEDGE", the flashing lamps shall go to continuously off if the ac power is not present at the time of acknowledge, or to continuously lit if the ac power has been restored at the time of acknowledge. Actuation of the switch to "ACKNOWLEDGE" shall at no time prohibit the receipt of any power status change from resulting in the operation described above. Actuation of the switch to the "LAMP TEST" position shall cause both lamps to be continuously lit while the switch is actuated.

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3.5.3 Primary power supply. The primary power supply shall operate as specified herein from 110 to 125 volts, 48 to 62 Hz ac power. The power supply shall provide a 20 ±1 Vdc regulated output voltage, a 22.5 ±3.5 volts ac (Vac) output voltage, and an unregulated dc voltage as specified herein to maintain a standby battery supply on trickle charge. The power supply shall also provide, at the battery terminals TB2-2 to TB2-1, a fast charge voltage which shall function as specified herein. The trickle charge voltage shall vary as a function of the applied loads and ambient temperature in accordance with table I and the Appendix A test procedures. The fast charge voltage output shall be capable of supplying up to the maximum current specified in table II. The fast charge mode shall be initiated when the voltage at the battery terminals falls to 24.0 ±0.3 Vdc (fast charge threshold voltage) for the 1 and 5 zone and 26.0 ±0.3 Vdc for the 25 zone power supply at a room ambient temperature of 77 ±5 °F. For temperatures outside this temperature range, the fast charge threshold voltage tolerance shall be ±1 Vdc. The trickle charge mode shall be restored when the battery terminal voltage rises 2 ±0.5 volts above the fast charge threshold voltage and shall exist whenever the battery terminal voltage is above the fast charge threshold voltage, provided fast charge operation has not been initiated. The power supply shall also conform to the current output, line regulation, load regulation and ripple voltage requirements specified in table II. A power off/on switch shall be provided which in the "OFF" position shall disconnect ac power and the standby battery power from the power supply circuitry. With the power switch in the "ON" position, a loss of input ac power shall cause the power supply to automatically operate from the standby battery supply without false alarm, incorrect status indication or other impairments of function. With a battery (batteries) or equivalent standby supply capable of providing the required current at 24 Vdc, the power supply shall continue to provide the regulated 20 Vdc output specified. The twenty-five zone power supply shall have a thermal overload switch (thermostat) in the battery fast charge circuit which shall open when the switch temperature rises to 150 ±5 °F and close when the temperature falls to 130 ±6 °F.

3.6 Burn-in. Each monitor cabinet (power supply component only) and power supply when furnished separately shall be subjected to an eight-hour burn-in to reduce the number of infant mortality failures (see 4.4.2.4).

TABLE I. Charging voltage as function of loads and temperatures.

Power Supply Type	**Applied Loads**			**Voltage: TB2-2 to TB2-1 (Vdc)**		
	TB3-1 to TB3-7	TB3-2 to TB3-7	TB2-2 to TB2-1	77 ±5 °F	-20 °F	+150 °F
1 Zone	no load	no load	no load	27.9 ±.3	30.2 ±.7	25.6 ±.7
1 Zone	20K ohms	80 ohms	270 ohms	27.0 ±.5	29.8 ±.8	25.2 ±.8
5 Zone	no load	no load	no load	28.2 ±.3	30.8 ±.7	26.6 ±.7
5 Zone	20K ohms	16 ohms	150 ohms	26.4 ±.8	29.0 ±1.0	24.8 ±1.0
25 Zone	no load	no load	no load	29.4 ±.8	32.8 ±1.0	27.0 ±1.0
25 Zone	20K ohms	3.2 ohms	no load	27.7 ±.3	31.0 ±.8	25.4 ±.8

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TABLE II. Power supply general requirements.

Characteristic	1 Zone	5 Zone	25 Zone
Maximum current			
TB3-2 to TB3-7	0.25 amperes (A)	1.25 A	6.25 A
TB3-1 to TB3-7	1.4 mA ac	1.4 mA ac	1.4 Ma ac
TB2-2 to TB2-1 (trickle charge)	0.25 A	0.300 A	0.750 A
TB2-2 to TB2-1 (fast charge)	2 A	4A	8A
Line Regulation			
TB3-2 to TB3-7	0.5 %	0.5 %	0.5 %
TB3-1 to TB3-7	10 %	10 %	10 %
TB2-2 to TB2-1	1.5 %	1.5 %	1.5 %
Load Regulation			
TB3-2 to TB3-7	1 %	1 %	1 %
TB3-1 to TB3-7	5 %	5 %	5 %
Ripple Content TB3-2 to TB3-7 (periodic & random variation)	400 millivolts (mV) peak to peak (P-P) at 0.25A	400 mV P-P at 1.25 A	400 mV P-P at 1.25 A

3.7 Reliability. When tested as specified in 4.4.2.8, the monitor cabinets shall have a design mean-time-between-failure (MTBF) of 10,000 hours or greater.

3.8 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustments, and without selection for fit or performance.

3.9 Identification marking. The monitor cabinets and major components shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

3.10 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.5) for manufacturing and testing monitor cabinets and major components when furnished separately:

Item No.	Description	Identification	Quantity
1.	Stable base drawings of printed wiring board (PWB), logic.	13220E3983, sh 2	1 1/
2.	Stable base drawing of PWB, power supply (1 zone).	13220E3990, sh 2	1 2/
3.	Stable base drawing of PWB, power supply (5 zone)	13220E4017, sh 1 and 2	1 per sheet 3/
4.	Stable base drawing of PWB, power supply (25 zone)	13220E4134, sh 2	1 4/

1/ Provided only when procuring any monitor cabinet and signal modules.
2/ Provided only when procuring 1 zone monitor cabinets and 1 zone power supplies.
3/ Provided only when procuring 5 zone monitor cabinets and 5 zone power supplies.
4/ Provided only when procuring 25 zone monitor cabinets and 25 zone power supplies.

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3.11 Workmanship. Workmanship shall be in accordance with the guidelines and guidance paragraphs of MIL-HDBK-454, guideline 9, or equivalent.

3.11.1 Welding. Sheet metal welding shall be in accordance with AWS D9.1, except no visible pores shall be allowed.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 Examination. Each first article monitor cabinet and major components when furnished separately shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.2.2 Tests.

4.2.2.1 Monitor cabinets. Monitor cabinets shall be subjected to the tests marked "X" in column 1 of table III, except as noted. Failure of any test shall be cause for rejection. Unless otherwise specified (see 6.2), five monitor cabinets shall be subjected to environmental tests (see 4.4.2.7) and twelve shall be subjected to reliability tests (see 4.4.2.8).

4.2.2.2 Major components when furnished separately. Major components when furnished separately shall be subjected to the tests marked "X" in column 1 of tables IV and V, as applicable, except as noted. Failure of any test shall be cause for rejection. Unless otherwise specified (see 6.2), the quantity of major components required for first article tests shall be five units, each of which shall be subjected to all tests listed in the applicable test schedule (see 4.4.2.2).

4.3 Conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with ANSI/ASQC Z1.4, inspection level II. Sampling for tests shall be in accordance with ANSI/ASQC Z1.4, inspection level S-2.

4.3.2 Examination.

4.3.2.1 Individual. Each monitor cabinet and major component shall be examined for the critical defect specified in 4.4.1. Presence of the defect shall be cause for rejection.

4.3.2.2 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major defects specified in 4.4.1. Sample size shall be determined by using ANSI/ASQC Z1.4, table I and table II-A. A lot shall be accepted when zero defects are found and rejected when 1 or more defects are found.

4.3.3 Tests.

4.3.3.1 Individual.

4.3.3.1.1 Monitor cabinets. Each monitor cabinet, shall be subjected to the tests marked "X" in column 2 of table III. Failure of any test shall be cause for rejection.

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4.3.3.1.2 Major components when furnished separately. Each major component, when furnished separately, shall be subjected to the tests marked "X" in column 2 of tables IV and V. Failure of any test shall be cause for rejection.

4.3.3.2 Samples.

4.3.3.2.1 Complete monitor cabinets. Each sample monitor cabinet, selected in accordance with 4.3.1, shall be subjected to the tests marked "X" in column 3 of table III.

4.3.3.2.2 Major components when furnished separately. Each sample major component when furnished separately, selected in accordance with 4.3.1, shall be subjected to the tests marked "X" in column 3 of tables IV or V, as applicable. Failure of any test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The monitor cabinets and the major components when furnished separately shall be examined as specified herein for the following defects (as applicable):

Critical.

1. Welding burn through.

Major.

101. Electromagnetic interference suppression components and networks not as specified on drawings (see 3.1 and 3.1.1).
102. Components missing or not as specified on drawings(see 3.1, 3.1.1, and 3.1.2).
103. Assembly not in accordance with drawings (see 3.1, 3.1.1, and 3.1.2).
104. Materials not as specified (see 3.3).
105. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
106. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
107. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the monitor cabinets or major components when furnished separately (see 3.3.3).
108. Interchangeability not as specified (see 3.8).
109. Identification or special marking missing, incomplete or illegible (see 3.9).
110. Workmanship not as specified (see 3.11).
111. Welding not in accordance with AWS D9.1 (see 3.11.1).

4.4.2 Tests.

4.4.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient humidity conditions and at a room ambient temperature of 77 ±5 °F. Unless otherwise specified herein, the monitor cabinets and major components when furnished separately shall operate as specified herein without maintenance.

4.4.2.2 Test schedule.

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TABLE III. Test schedule for complete monitor cabinets.

First article	Quality conformance		Test 1/	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	–	Continuity 4/	3.5.1	4.4.2.5
X	X	–	Burn-in 5/	3.6	4.4.2.4
X	X	–	Functional	3.5.2, 3.5.3	4.4.2.6.1
X 2/	–	X	Low temperature	3.4.1(a), 3.4.2(a)	4.4.2.7.1
X 2/	–	X	High temperature	3.4.1(a), 3.4.2(a)	4.4.2.7.2
X 2/	–	X	Humidity	3.4.1(b), 3.4.2(b)	4.4.2.7.3
X 2/	–	–	Shock	3.4.2(c)	4.4.2.7.4
X 3/	–	–	Reliability	3.7	4.4.2.8

1/ To test, power supplies and signal modules may be removed from the monitor cabinets and subjected to required tests individually (except for continuity test).
2/ Reliability test units not subjected to this test.
3/ Environmental test units not subjected to this test.
4/ Portions of continuity test involving power supply and signal module connections must be performed or repeated after completing all other tests.
5/ Burn-in required for power supply component only.

TABLE IV. Test schedule for power supply.

First article	Quality conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	–	Burn-in	3.6	4.4.2.4
X	X	–	Functional	3.5.3	4.4.2.6.2
X	–	X	Low temperature	3.4.1(a), 3.4.2(a)	4.4.2.7.1
X	–	X	High temperature	3.4.1(a), 3.4.2(a)	4.4.2.7.2
X	–	X	Humidity	3.4.1(b), 3.4.2(b)	4.4.2.7.3
X	–	–	Shock	3.4.2(c)	4.4.2.7.4

TABLE V. Test schedule for signal module.

First article	Quality conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	–	Functional	3.5.2	4.4.2.6.3
X	–	X	Low temperature	3.4.1(a), 3.4.2(a)	4.4.2.7.1
X	–	X	High temperature	3.4.1(a), 3.4.2(a)	4.4.2.7.2
X	–	X	Humidity	3.4.1(b), 3.4.2(b)	4.4.2.7.3
X	–	–	Shock	3.4.2(c)	4.4.2.7.4

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4.4.2.3 Test methods. Test shall be conducted in accordance with the test methods in the appendices and as specified herein. Calibration of the test equipment shall be in accordance with ISO 10012-1.

4.4.2.4 Burn-in. Each monitor cabinet (power supply component only) and each power supply when furnished separately shall be subjected to an eight-hour burn-in. Power supplies shall be aligned in accordance with the Appendix B alignment procedure prior to the functional test. The appropriate functional test of 4.4.2.6 shall be performed after completion of burn-in. During the burn-in, each unit shall be subjected to an abbreviated test as specified herein three times, but no test shall be initiated less than two hours after completion of the previous test period. Except as specified during the test periods:

a. Each power supply under test shall be continuously energized by the primary power source specified;
b. Each power supply, whether as a component of a monitor cabinet, or when furnished separately, shall have its 20 Vdc and 22.5 Vac outputs deliver maximum rated currents (±5 percent) specified in table II.

The test to be performed three times during the eight-hour burn-in shall be to introduce the fast charge load specified in the test method appropriate to the power supply being tested and maintain for 5 ±1 seconds; after restoring to float charge operation, remove load from 20 volt output for 5 ±1 seconds; for each change of operation, verify the voltages during and after the changes are within allowable tolerances. Noncompliance with the applicable portions of 3.5 shall constitute failure. Failures during burn-in shall be analyzed, repaired, and corrective action taken as necessary. When specified in the contract or purchase order (see 6.2), burn-in test reports shall be forwarded to the contracting officer. Any unit failing during burn-in shall be subjected to an additional eight-hour burn-in after failure analysis and repair and successful completion of the functional test.

4.4.2.5 Continuity. Each monitor cabinet shall be tested in accordance with Appendix A, test method 1. Nonconformance to 3.5.1 shall constitute failure of this test.

4.4.2.6 Functional. Functional tests of the monitor cabinet and major components when furnished separately shall be in accordance with 4.4.2.6.1 through 4.4.2.6.3 as applicable.

4.4.2.6.1 Monitor cabinet. The 1, 5 , and 25 zone monitor cabinets shall be tested by performing the functional test for the power supply in accordance with Appendix A, test method 2, 3, or 4 as applicable and the functional test for the signal module in accordance with Appendix A, test method 5. To perform those tests, the power supply and signal module may be removed from the monitor cabinet. Nonconformance to 3.5.2 and 3.5.3 shall constitute failure of this test.

4.4.2.6.2 Power supply. The power supply shall be tested in accordance with Appendix A, test methods 2, 3, or 4 as applicable. Nonconformance to 3.5.3 shall constitute failure of this test.

4.4.2.6.3 Signal module. The signal module shall be tested in accordance with Appendix A, test method 5. Nonconformance to 3.5.2 shall constitute failure of this test.

4.4.2.7 Environmental. The monitor cabinets and major cabinets when furnished separately shall be subjected to the following environmental tests.

4.4.2.7.1 Low temperature. Low temperature test shall be conducted in accordance with the low temperature test of Appendix C.

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4.4.2.7.2 High temperature. High temperature test shall be conducted in accordance with the high temperature test of Appendix C.

4.4.2.7.3 Humidity. Humidity test shall be conducted in accordance with the humidity test of Appendix C.

4.4.2.7.4 Shock. Shock test (bench handling) shall be performed in accordance with the shock test of Appendix C.

4.4.2.7.5 Failure criteria. Nonconformance to 3.4 shall constitute failure of these tests.

4.4.2.8 Reliability. Using an upper test MTBF (θ_0) of 10,000 hours and a lower test MTBF (θ_1) of 5,000 hours, the monitor cabinets shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-HDBK-781, test plan VIIID. Continue testing until either an "accept" or "reject" decision is reached. All test incidents will be scored by the Government. A failure is defined as any malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
b. Serious damage (see 6.3.2) to the system.
c. Serious personnel safety hazards.

4.4.2.8.1 Test procedures. One or more monitor cabinets shall be randomly selected by a Government representative. The tests shall be conducted under ambient humidity conditions at a room temperature of 77 ±5 °F. Equipment shall be energized at the primary power input specified herein. Load resistors R3 and R4 of the applicable Appendix A test methods (2, 3, or 4) shall be placed across the 20 Vdc and 27 Vdc outputs respectively, except as required during the test periods described below. Operation shall be continuous throughout the test. Nine times each 24-hours, but not more often than once per hour, complete functional tests equivalent to those of Appendix A test methods 2, 3, or 4 (as applicable) and 5 shall be performed, except tests to determine sound pressure level, line and load regulation, and fast charge cut-in and cut-out thresholds, and, for the 25 zone, the thermal cut-off and external input power, need to be performed only once daily. The test shall commence only after successful completion of functional testing and a burn-in period of eight hours (power supply only), as specified in 4.4.2.4. Failed indicator lamps shall be noted as to time failure is detected and shall be replaced. Such failures shall not be counted as relevant and any dependent failures caused thereby are counted as relevant. Alarms not deliberately excited as part of the equipment test routine shall be reported and cause of the alarm identified, if practicable. This accounting shall be maintained separately from the failure accounting. No adjustments shall be allowed during the test.

4.4.2.8.2 Failure criteria. Nonconformance to 3.7 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The monitor cabinet is intended for use as the alarm and status display component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. Nomenclature and quantity of item(s) to be furnished (see 1.2, 3.1 and 3.1.1).
d. Technical Data Package List(s), including revision level, to which each item being furnished (by contractor) is to be built.
e. When a first article is required for inspection and approval, the number of units required, and the allocation of required units to environmental and reliability tests respectively, if other than specified (see 3.2, 4.2.2.1, and 4.2.2.2).
g. When Government-loaned property is not required (see 3.10).
h. Schedule, content, and format of burn-in test reports (see 4.4.2.4).
i. Packaging requirements (see 5.1).

6.3 Definitions. The following definitions apply throughout this specification.

6.3.1 False alarm. An alarm generated by stimuli other than those for which the test item is designed to alarm.

6.3.2 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any components; physical change which impairs the mechanical integrity of the monitor cabinets; evidence of delamination or water penetration into integrated circuits, printed circuit boards, or external components, or physical condition resulting in noncompliance of the item with the provisions of this specification.

6.3.3 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the objectives of the product and assure interchangeability of the repair parts.

6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the end items.

6.5 Government-loaned property. The contracting officer should arrange to lend the property specified in 3.10 when applicable (see 6.2).

6.6 Subject term (key word listing).

Monitor unit
Intrusion detection
Physical security
Alarm system
J-SIIDS

6.7 Quality assurance provisions (QAP) records. The contracting officer should include a clause requiring the contractor to maintain complete records of all examinations and tests performed to verify the requirements of

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classified QAP characteristics. Such clause should also require that the records include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR4
Navy – SH
Air Force – 99

Preparing activity:
Army – CR4

Project 6350-0209

Review activities:
Army – CR
Navy – CG, MC
DLA – GS

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APPENDIX A

TEST METHODS

FOR

MONITOR CABINETS AND MAJOR COMPONENTS

A.1 SCOPE

A.1.1 Scope. The test methods contained in this appendix are to determine whether monitor cabinets and major components furnished separately, procured under this specification, conform to the requirements specified. Figures 1 through 4 contain data for fabrication of test fixtures and preparation of test set-ups which are required to facilitate testing of the monitor cabinets and major components furnished separately. Power supply alignment procedures are provided in Appendix B. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

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TEST METHOD 1

MONITOR CABINET CONTINUITY TEST

1. TEST APPARATUS: Ohmmeter.

2. TEST PROCEDURE:

2.1 Using an ohmmeter, check continuity between all connections listed on the wire list of the top assembly drawing for the applicable monitor cabinet (13220E3900 for the type A, 13220E4000 for the type B, 13220E4100 for the type C monitor cabinets), except the connections listed in step 2.2 below, which shall be verified after all other tests have been completed and the power supply and signal module have been installed in the monitor cabinet. Other means to verify continuity may be used if approved by contracting officer.

2.2 After all other tests have been completed and the power supply (PS) and signal module (SM) have been installed in the monitor cabinet, verify continuity between the following:

Type A monitor cabinet:

PS TB2-2 to battery #1 + lead	PS TB3-4 to J1-9
PS TB2-2 to battery #2 + lead	PS TB3-6 to J1-10
PS TB2-1 to battery #1 - lead	SM TB1-3 to J1-9
PS TB2-3 to battery #2 - lead	SM TB1-4 to J1-10
PS TB3-1 to SM TB1-2	SM TB1-1 to J1-14
PS TB3-2 to PS TB3-3 and PS TB3-4	PS TB3-5 to PS TB3-6 and PS TB3-7

Type B monitor cabinet:

PS TB2-2 to battery + lead	PS TB3-3 to SM TB1-3
PS TB2-3 to battery - lead	PS TB3-4 to J5-9
PS TB2-2 to PS TB2-4	PS TB3-5 to J5-10
PS TB2-3 to PS TB2-1	PS TB3-6 to SM TB1-4
PS TB3-1 to SM TB1-2	SM TB1-1 to J1-14
PS TB3-2 to PS TB3-3 and PS TB3-4	PS TB3-5 to PS TB3-6 and PS TB3-7

Type C monitor cabinet:

PS TB2-1 to battery #1 - lead	PS TB3-3 to J25-9
PS TB2-2 to battery #1 + lead	PS TB3-5 to J5-10
PS TB2-3 to battery #2 - lead	PS TB3-3 to J10-10
PS TB2-4 to battery #2 + lead	PS TB3-6 to J15-10
PS TB2-5 to battery #3 - lead	PS TB3-6 to J20-10
PS TB2-6 to battery #3 + lead	PS TB3-6 to J25-10
PS TB3-1 to SM TB1-2	SM TB1-3 to J1-9
PS TB3-2 to J5-9	SM TB1-4 to J1-10
PS TB3-2 to J10-9	SM TB1-1 to J1-14
PS TB3-3 to J15-9	PS TB4-1 to Ext. DC +
PS TB3-3 to J20-9	PS TB4-2 to Ext. DC -
PS TB3-2 to PS TB3-3 and PS TB3-4	PS TB3-5 to PS TB3-6 and PS TB3-7

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TEST METHOD 2

1 ZONE POWER SUPPLY

FUNCTIONAL TESTS

1. TEST APPARATUS:

a.	Oscilloscope (scope)	Tektronix model 2225 or equivalent.
b.	Voltmeters, dc (M2, M3, M4)	3-1/2 digit or greater, minimum accuracy 0.25 percent of reading ± 1 least significant digit (LSD).
c.	Voltmeters, ac (M1, M5)	3-1/2 digit or greater, minimum accuracy 0.25 percent of reading ± 1 LSD.
d.	Power supply (PS1)	Hewlett-Packard Model 6291A or equivalent.
e.	Autotransformer (VR1)	General Radio Model W20M or equivalent.
f.	SPST switches (S1-S7)	Leviton Model 19916 or equivalent.
g.	Resistor, 20 Kilohm (R1)	20K ohm ±5 percent, 1/4 watt.
h.	Resistor, 160 ohm (R2)	Ohmite, 200 ohm, adjustable, 12 watts, type 210-12, P/N D12K200 or equivalent.
i.	Resistor, 80 ohm (R3)	Ohmite, 100 ohm, adjustable, 12 watts, type 210-12, P/N D12K100 or equivalent.
j.	Resistor, 270 ohm (R4)	Ohmite, 300 ohm, adjustable, 12 watts, type 210-12, P/N D12K300 or equivalent.
k.	Resistor, 15 ohm (R5)	Ohmite, 25 ohm, adjustable, 175 watts, type 210-175, P/N D175K25R or equivalent.
l.	Ammeter, dc (see note)	3-1/2 digit or greater, minimum accuracy 0.75 percent of reading ± 1 LSD, 4 amperes minimum.
m.	Test fixture.	(See figure 1).

NOTE: Item l. is required only for calibrating test fixture resistors.

2. TEST PROCEDURE:

2.1 When this test is being performed as part of the monitor cabinet functional test, remove the power supply from the cabinet prior to performing the test procedure.

3. TEST PROCEDURE

3.1 Output voltage levels.

a. Connect the power supply under test (PSUT) in the test fixture as shown in figure 1. Set the PSUT power switch in the "OFF" position and apply 117 Vac, 60 Hz line power to the autotransformer (VR1).
b. Open switches S1 through S7.
c. Adjust the VR1 for 117 Vac on M5.
d. Close S4.
e. Place the PSUT power switch in the "ON" position. Allow at least 15 minutes warm-up before proceeding with test.
f. The following output voltage levels shall be observed:

Output	Location	Meter	Voltage
1	TB3-2 to TB3-5	M2	20 ±1 Vdc
2	TB2-2 to TB2-1	M3	See below
3	TB3-1 to TB3-6	M1	22.5 ±3.5 Vac

Output 2: 27.9 ±0.3 Vdc at 77 ±5 °F, 30.2 ±0.7 Vdc at -20 °F, 25.6 ±0.7 Vdc at +150 °F, and 27.5 ±0.7 Vdc during humidity test.

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g. Close S1, S3, and S6 and record voltages on M2, M3, and M1. M2 and M1 voltages shall be as in step f. M3 voltage shall be 27.0 ±0.5 Vdc at 77 ±5 °F, 29.8 to ±0.8 Vdc at -20 °F, 25.2 ±0.8 Vdc at +150 °F, and 26.7 ±0.8 Vdc during humidity test.
h. The output voltages must be within the above levels to be acceptable.

3.2 Line regulation.

a. Close S1, S3, and S4 and open S2, S5, S6, and S7.
b. Place the power supply under test power switch in the "ON" position.
c. Adjust the VR1 for 117 Vac reading on M5.
d. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-2 to TB3-5	M2
2	TB2-2 to TB2-1	M3
3	TB3-1 to TB3-6	M1

e. Adjust the VR1 for 125 Vac.
f. Repeat step d.
g. Adjust the VR1 for 110 Vac.
h. Repeat step d.
i. For acceptable line regulation, the output voltage measurements taken in steps f. and h. shall not differ from those taken in step d. by more than the following:

Output 1: ±0.5 percent.
Output 2: ±1.5 percent.
Output 3: ±10 percent.

3.3 Load regulation.

a. Close S4 and S6 and open S1, S2, S3, S5, and S7.
b. Adjust VR1 for 117 Vac as indicated on M5.
c. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-2 to TB3-5	M2
3	TB3-1 to TB3-6	M1

d. Close S1 and S3. Repeat the voltage readings as in step c.
e. Open S3 and close S2. Repeat the voltage readings as in step c.

f. For acceptable load regulation, the output voltage measurements taken in steps d. and e. shall not differ from those taken in step c. by more than the following:

Output 1: ±1 percent
Output 2: ±5 percent.

3.4 Ripple and noise (periodic and random deviations).

a. Open S2, S5, and S7 and close S1, S3, S4, and S6.
b. Measure the output ac P-P voltage at output 1 (TB3-2 to TB3-5) with the scope. This voltage shall include spikes caused by the switching regulator and any other source. The intensity on the scope may need to be turned up to observe these spikes. The lead lengths from the scope to TB3 should be as short as possible.
c. For acceptable ripple and noise, including spikes caused by the switching regulator, the ac P-P voltage shall be less than 400 mV.

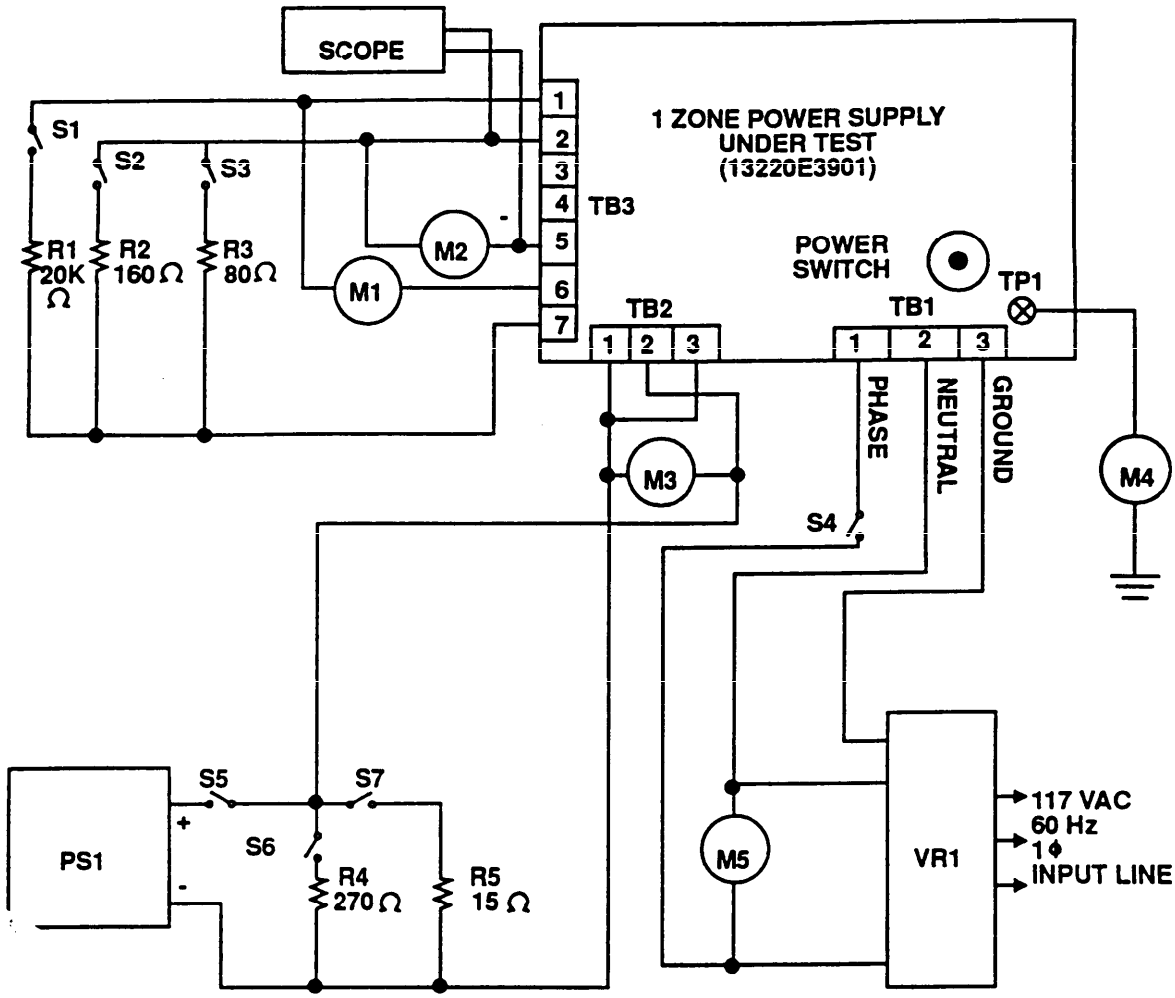
3.5 Battery charger interface characteristics.

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NOTE: During low temperature, high temperature and humidity tests, steps b through d below need not be performed.

a. Open test fixture switches S1 through S7 and adjust PS1 for 27 Vdc output.
b. Monitor TP1 on PWB A3 (meter M4). Close test fixture switch S5 and power switch on PSUT. Meter M4 shall indicate 9 ±6 Vdc.
c. Slowly decrease PS1 voltage while observing M3 and M4. Record the voltage on M3 when the voltage on M4 suddenly decreases to less than 1 Vdc. M3 voltage shall be 24.0 ±0.3 Vdc.
d. Slowly increase PS1 voltage until M4 voltage returns abruptly to 9 ±6 Vdc. Record voltage on M3 when the change occurs. It shall be 2.0 ±0.5 Vdc greater than the voltage of step c. above.
e. Repeat steps c. and d. Values shall be within 0.2 Vdc of those previously obtained. If not, repeat, changing PS1 voltages very slowly until repeatable results are obtained.
f. Decrease PS1 voltage to 23.8 ±0.2 V and close S3. Observe voltage on M2. It shall be 20 ±1 Vdc.
g. Open S5 and close S4. Remove oscilloscope probe from across TB3-2 and TB3-5 and place across TB2-2 and TB2-1.
h. Momentarily close S7 while observing voltage on scope. The voltages shall fluctuate between 2 values while the fast charge circuit is activated, 38 ±5 V and 13 ±7 V. Record the values.
i. For acceptance, the battery charger interface characteristics shall be within the above specified limits.
j. After completion of all tests, reinstall power supply in cabinet, if applicable, and verify proper connections by performing step 2.2 of test method 1.

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CALIBRATION PROCEDURE

To calibrate fixture, remove 1 Zone supply. Using a regulated power supply and an ammeter, calibrate each resistor in turn as follows (assume terminal strips labelled and numbered as shown, but without the power supply under test):

R3: With S3 closed, S2 open and an ammeter between the power supply and the lead going to TB3-2, set R3 for 20.0 Vdc on M2 with current of 250 mA (+/- 3%).

R2: With S2 closed, S3 open and an ammeter between the power supply and the lead going to TB3-2, set R2 for 20.0 Vdc on M2 with current of 125 mA (+/- 3%).

R4: With S6 closed, S5 and S7 open and an ammeter between the power supply and the lead going to TB2-2, set R4 for 27.0 Vdc on M3 with current of 100 mA (+/- 3%).

R5: With S7 closed, S5 and S6 open and an ammeter between the power supply and the lead going to TB2-2, set R5 for 30.0 Vdc on M3 with current 2.0 A (+/- 3%).

R1: With S1 closed and an ammeter between the power supply and the lead going to TB3-1, select R1 for 22.5 Vac from TB3-1 to TB3-6 with current 1.125 mA (+/- 3%).

NOTES: a. Tolerances are 3% for current, 1% for voltage.
b. Negative lead of M4 shall be connected to any circuit ground.

FIGURE 1. Test fixture for 1 zone power supply.

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TEST METHOD 3

5 ZONE POWER SUPPLY

FUNCTIONAL TESTS

1. TEST APPARATUS:

a.	Oscilloscope (Scope)	Tektronix model 2225 or equivalent.
b.	Voltmeters, dc (M2, M3, M4)	3-1/2 digit or greater, minimum accuracy 0.25 percent of reading ± 1 LSD.
c.	Voltmeter, ac (M1, M5)	3-1/2 digit or greater, minimum accuracy 1 percent of reading ± 1 LSD.
d.	Power supply (PS1)	Hewlett-Packard Model 6291A or equivalent.
e.	Autotransformer (VR1)	General Radio Model W20M or equivalent.
f.	SPST switches (S1-S7)	Leviton Model 19916 or equivalent.
g.	Resistor, 20 Kilohm (R1)	20K ohms, ±5 percent, 1/4 watt.
h.	Resistor, 32 ohm (R2)	Ohmite, 50 ohm, adjustable, 50 watts, type 210-50, P/N D50K50R or equivalent.
i.	Resistor, 16 ohm (R3)	Ohmite, 25 ohm, adjustable, 50 watts, type 210-50, P/N D50K25R or equivalent.
j.	Resistor, 150 ohm (R4)	Ohmite, 200 ohm, adjustable, 12 watts, type 210-12, P/N D12K200 or equivalent.
k.	Resistor, 7.5 ohm (R5)	Ohmite, 10 ohm, adjustable, 225 watts, type 210-225, P/N D225K10R or equiv.
l.	Ammeter, dc (see note)	3-1/2 digit or greater, minimum accuracy 0.75 percent of reading ± 1 LSD, 4 amperes minimum.
m.	Test fixture	(See figure 2).

Note: Item l. is required only for calibrating test fixture resistors.

2. TEST PREPARATION:

2.1 When this test is being performed as part of the monitor cabinet functional test remove the power supply from the cabinet prior to performing the test procedure.

3. TEST PROCEDURE

3.1 Output voltage levels.

a. Connect the power supply under test (PSUT) in the test fixture as show in figure 2. Set PSUT power switch in the "OFF" position and apply 117 Vac, 60 Hz line power to the autotransformer (VR1).
b. Open S1 through S7.
c. Adjust the VR1 for 117 Vac on M5.
d. Close S4.
e. Place the power supply under test power switch in the "ON" position., Allow at least 15 minutes warm up before proceeding with test.
f. The following output voltage levels shall be observed:

Output	Location	Meter	Voltage
1	TB3-2 to TB3-5	M2	20 ±1 Vdc
2	TB2-4 to TB2-1	M3	See below
3	TB3-1 to TB3-6	M1	22.5 ±3.5 Vac

Output 2: 28.2 ±0.3 Vdc at 77 ±5 °F; 30.8 ±0.7 Vdc at-20 °F; 26.6 +0.7 Vdc at +150 °F; 27.9 ±0.5 Vdc during humidity test.

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g. Close S1, S3, and S6 and record voltages on M2, M3, and M1. M2 and M1 voltages shall be as in step f. M3 voltage shall be 26.4 ±0.8 at 77 ±5 °F, 29.0 ±1.0 at -20 °F, 24.8 ±1.0 at +150 °F, and 26.1 ±1.0 Vdc during humidity test.
h. The output voltages must be within the above levels to be acceptable.

3.2 Line regulation.

a. Close S1, S3, and S4 and open S2, S5, S6, and S7.
b. Place the power supply under test power switch in the "ON" position.
c. Adjust the VR1 for 117 Vac reading on M5.
d. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-2 to TB3-5	M2
2	TB2-4 to TB2-1	M3
3	TB3-1 to TB3-6	M1

e. Adjust the VR1 for 125 Vac.
f. Repeat step d.
g. Adjust the VR1 for 110 Vac.
h. Repeat step d.
i. For acceptable line regulation, the output voltage measurements taken in steps f. and h. shall not differ from those taken in step d. by more than the following:

Output 1: ±0.5 percent.
Output 2: ±1.5 percent.
Output 3: ±10 percent.

3.3 Load regulation.

a. Close S4 and S6 and open S1, S2, S3, S5, and S7.
b. Adjust VR1 for 117 Vac as indicated on M5.
c. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-2 to TB3-5	M2
3	TB3-1 to TB3-6	M1

d. Close S1 and S3. Repeat the voltage readings as in step c.
e. Open S3 and close S2. Repeat the voltage readings as in step c.
f. For acceptable load regulation, the output voltage measurements taken in steps d. and e. shall not differ from those taken in step c. by more than the following:

Output 1: ±1 percent.
Output 3: ±5 percent.

3.4 Ripple and noise (periodic and random deviations).

a. Open S2, S5, and S7 and close S1, S3, S4, and S6.
b. Measure the output ac P-P voltage at output 1 (TB3-2 to TB3-5) with the scope. This voltage shall include spikes caused by the switching regulator and any other source. The intensity on the scope may need to be turned up to observe these spikes. The lead lengths from the scope to TB3 should be as short as possible.
c. For acceptable ripple and noise, including spikes caused by the switching regulator, the ac P-P voltage shall be less than 400 mV.

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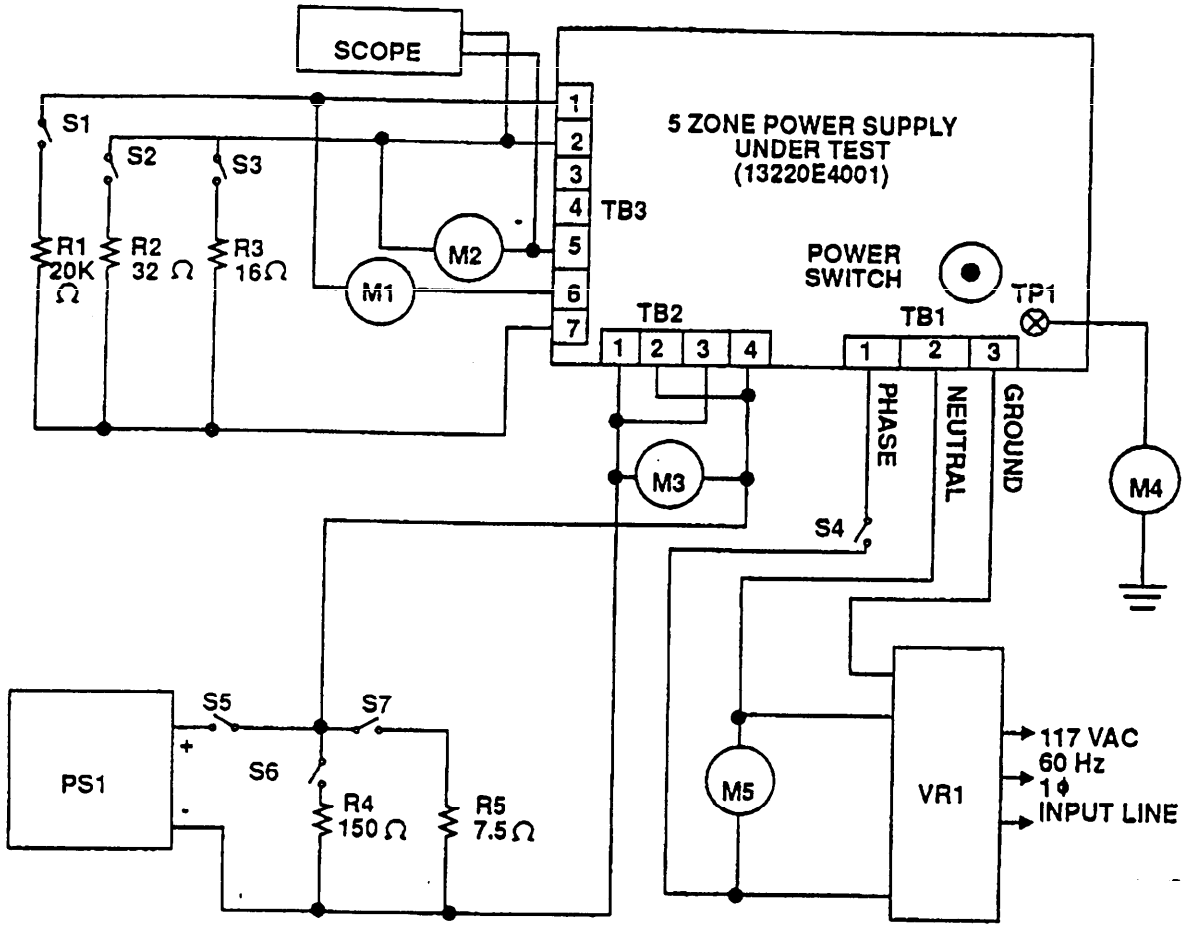
3.5 Battery charger interface characteristics.

NOTE: During low temperature, high temperature and humidity tests, steps b. through d. below need not be performed.

a. Open test fixture switches S1 through S7 and adjust PS1 for 27 Vdc output.
b. Monitor TP1 on PWB A1 (meter M4). Close test fixture switch S5 and power switch on PSUT. Meter M4 shall indicate 9 ±6 Vdc.
c. Slowly decrease PS1 voltage while observing M3 and M4. Record the voltage on M3 when the voltage on M4 suddenly decreases to less than 1 Vdc. M3 voltage shall be 24.0 ±0.3 Vdc.
d. Slowly increase PS1 voltage until M4 voltage returns abruptly to 9 ±6 Vdc. Record voltage on M3 when the change occurs. It shall be 2.0 ±0.5 Vdc greater than the voltage of step c. above.
e. Repeat steps c. and d. Values shall be within 0.2 Vdc of those previously obtained. If not, repeat, changing the power supply PS1 voltages very slowly until repeatable results are obtained.
f. Decrease PS1 voltage to 23.8 ±0.2 V and close S3. Observe voltage on M2. It shall be 20 ±1 Vdc.
g. Open S5 and close S4. Remove oscilloscope probe from across TB3-2 and TB3-5 and place across TB2-4 and TB2-1.
h. Momentarily close S7 while observing voltage on scope. The voltages shall fluctuate between 2 values while the fast charge circuit is activated, 38 ±5 V and 13 ±7 V. Record the values.
i. For acceptance, the battery charger interface characteristics shall be within the above specified limits.
j. After completion of all tests, reinstall power supply in cabinet, if applicable, and verify proper connections by performing step 2.2 of test method 1.

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TEST METHOD 3 - Continued



CALIBRATION PROCEDURE

To calibrate fixture, remove 5 Zone supply. Using a regulated power supply and an ammeter, calibrate each resistor in turn as follows (assume terminal strips labelled and numbered as shown, but without the power supply under test):

R3: With S3 closed, S2 open and an ammeter between the power supply and the lead going to TB3-2, set R3 for 20.0 Vdc on M2 with current of 1.25 A (+/- 3%).

R2: With S2 closed, S3 open and an ammeter between the power supply and the lead going to TB3-2, set R2 for 20.0 Vdc on M2 with current of .625 A (+/- 3%).

R4: With S6 closed, S5 and S7 open and an ammeter between the power supply and the lead going to TB2-4, set R4 for 27.0 Vdc on M3 with current of 180 mA (+/- 3%).

R5: With S7 closed, S5 and S6 open and an ammeter between the power supply and the lead going to TB2-4, set R5 for 30.0 Vdc on M3 with current of 4.0 A (+/- 3%).

R1: With S1 closed and an ammeter between the power supply and the lead going to TB3-1, select R1 for 22.5 Vdc from TB3-1 to TB3-6 with current 1.125 mA (+/- 3%).

NOTES: a. Tolerances are 3% for current, 1% for voltage.
b. Negative lead of M4 shall be connected to any circuit ground.

FIGURE 2. Test fixture for 5 zone power supply

X-5040

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TEST METHOD 4

25 ZONE POWER SUPPLY

FUNCTIONAL TESTS

1. TEST APPARATUS

a. Oscilloscope (Scope)	Tektronix model 2225 or equivalent.
b. Voltmeters, dc (M2, M3, M4, M6)	3-1/2 digit or greater, minimum accuracy 0.25 percent of reading ± 1 LSD.
c. Voltmeter, ac (M1, M7)	3-1/2 digit or greater, minimum accuracy 1 percent of reading ± 1 LSD.
d. Power supply (PS1, PS2)	Hewlett-Packard Model 6267B or equivalent.
e. Power supply	Hewlett-Packard Model 6291A or equivalent.
f. Autotransformer (VR1)	General Radio Model W20M or equivalent.
g. SPST switches (S1-S9)	Leviton Model 19916 or equivalent.
h. Resistor, 20K ohm (R1)	20K ohms, ±5 percent, 1/4 watt.
i. Resistor, 6.4 ohm (R2)	Ohmite, 10 ohm, adjustable, 175 watts, type 210-175, P/N D175K10R or equiv.
j. Resistor, 3.2 ohm (R3)	Ohmite, 4 ohm, adjustable, 225 watts, type 210-225, P/N D225K4R0 or equiv.
k. Resistor, 72 ohm (R4)	Ohmite, 75 ohm, adjustable, 25 watts, type 210-25, P/N D25K75R or equivalent.
l. Resistor, 3.75 ohm (R5A, R5B)	Ohmite, 2 ohm, adjustable, 225 watts, type 210-225, P/N D225K2R0 or equiv.
m. Resistor, 14 ohm (R6)	Clarostat, 15 ohm, adjustable, 200 watts, type VK200WA or equivalent.
n. Thermometer	Hewlett-Packard, P/N HP2802A or equivalent.
o. Probe	Hewlett-Packard, P/N HP18640 PRT or equivalent.
p. Heat Gun	Ungar, P/N 6966C, or equivalent.
q. Ammeter, dc	3-1/2 digit or greater, minimum accuracy 1.5 percent of reading ± 1 LSD, 10 amperes minimum.
r. Test fixture	(See figure 3).

NOTES: 1. Power supply such as Hewlett-Packard model 6267B is required for calibrating test fixture resistors.
2. Item q. is required only for calibrating test fixture resistors.

2. TEST PREPARATION.

2.1 When this test is being performed as part of the monitor cabinet functional test, remove the power supply from the cabinet before performing the test procedure.

3. TEST PROCEDURE

3.1 Output voltage levels.

a. Connect the PSUT in the test fixture as shown in figure 3. Set power supply under test power switch in the "OFF" position and apply nominal 117 Vac, 60 Hz line power to the autotransformer (VR1).
b. Open S1 through S9.
c. Adjust the VR1 for 117 Vac on M5.
d. Close S4.
e. Place the PSUT power switch in the "ON" position. Allow at least 15 minutes warm-up before proceeding with test.

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TEST METHOD 4 - Continued.

f. The following output voltage levels shall be observed:

Output	Location	Meter	Voltage
1	TB3-3 to TB3-5	M2	20 ±1 Vdc
2	TB2-4 to TB2-3	M3	See below
3	TB3-1 to TB3-6	M1	22.5 ±3.5 Vac

Output 2: 29.4 ±0.8 Vdc at 77 ±5 °F, 32.8 ±1.0 Vdc at -20 °F, 27.0 ±1.0 Vdc at +150 °F, and 29.0 ±1.0 °F during humidity test.

g. Close S1 and S3 and record voltages on M2, M3 and M1. M2 and M1 voltages shall be as in step f. M3 voltage shall be 27.7 ±0.3 Vdc at 77 ±5 °F. 31.0 ±0.8 at -20 °F, 25.4 ±0.8 at +150 °F, and 27.7 ±1.0 Vdc during humidity test.
h. The output voltage must be within the above levels to be acceptable.

3.2 Line regulation.

a. Close S1, S3, and S4 and open S2, S5, S6, S7, S8, and S9.
b. Place the power supply under test power switch in the "ON" position.
c. Adjust the VR1 for 117 Vac reading on M5.
d. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-2 to TB3-5	M2
2	TB2-4 to TB2-3	M3
3	TB3-1 to TB3-6	M1

e. Adjust the VR1 for 125 Vac.
f. Repeat step d.
g. Adjust the VR1 for 110 Vac.
h. Repeat step d.
i. For acceptable line regulation, the output voltage measurements taken in steps f. and h. shall not differ from those taken in step d. by more than the following:

Output 1: ±1 percent.
Output 2: ±1.5 percent.
Output 3: ±10 percent.

3.5 Load regulation.

a. Close S4 and S6 and open all other test fixture switches.
b. Adjust VR1 for 117 Vac as indicated on M5.
c. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-3 to TB3-5	M2
3	TB3-1 to TB3-6	M1

d. Close S1 and S3. Repeat the voltage readings as in step c.
e. Open S3 and close S2. Repeat the voltage readings as in step c.
f. For acceptable load regulation, the output voltage measurements taken in steps d. and e. shall not differ from those taken in step c. by more than the following:

Output 1: ±1 percent.
Output 3: ±5 percent.

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TEST METHOD 4 - Continued.

3.4 Ripple and noise (periodic and random deviations).

a. Open S2, S5, S7, S8, and S9 and close S1, S3, S4, and S6.
b. Measure the output ac P-P voltage at output 1 (TB3-3 to TB3-5) with the scope. This voltage shall include spikes caused by the switching regulator and any other source. The intensity on the scope may need to be turned up to observe the spikes. The lead lengths from the scope to TB3 should be as short as possible.
c. For acceptable ripple and noise, including spikes caused by the switching regulator, the ac P-P voltage shall be less than 400 mV.

3.5 Battery charger interface characteristics.

NOTE: During low temperature, high temperature and humidity tests, steps b. through d. below need not be performed.

a. Open test fixture switches S1 through S9 and adjust PS1 for 29 Vdc output (meter M3).
b. Monitor TP1 on PWB A1 with meter M4 and close test fixture switch S5 and power switch on PSUT. Meter M4 shall indicate 9 ±6 Vdc.
c. Slowly decrease PS1 voltage while observing M3 and M4. Record the voltage on M3 when the voltage on M4 suddenly decreases to less than 1 Vdc. M3 voltage shall be 26.0 ±0.3 Vdc.
d. Slowly increase PS1 voltage until M4 voltage returns abruptly to 9 ±6 Vdc. Record voltage on M3 when change occurs. It shall be 2.0 ±0.5 Vdc greater than the voltage of step c. above.
e. Repeat steps c. and d. Values shall be within 0.2 Vdc of those previously obtained. If not, repeat, changing the power supply PS1 voltages very slowly until repeatable results are obtained.
f. Decrease PS1 voltage to 23.8 ±0.2 V and close S3. Observe voltage on M2. It shall be 20 ±1 Vdc.
g. Open S5 and close S4. Place scope probe across TB2-4 and TB2-1.
h. Momentarily close S7 while observing voltage on scope. The voltages shall fluctuate between 2 values while the fast charge circuit is activated, 35 ±5 V and 14 ±4 V. Record the values.
i. For acceptance, the battery charger interface characteristics shall be within the above specified limits.

3.6 Thermal cutoff.

NOTES:
1. Power supply cover is to be removed for this test.
2. This test need not be performed during low temperature, high temperature, and humidity testing.

a. Open S1, S2, S3, S5, S6, S7, and S9. Close S4 and S8.
b. Connect a thermometer via a temperature sensing probe to the heat sink in close proximity to the thermal switch (S2 in PSUT).
c. Place PSUT power switch in the "ON" position and adjust VR1 for 117 Vac on M5.
d. Place an oscilloscope across TB2-4 and TB2-1. Observe voltage on oscilloscope. Voltage shall fluctuate between 38 ±10 Vdc and 20 ±5 Vdc. Record values.
e. Direct heat gun (heat mode) to blow on thermal switch of PSUT.
f. As soon as the thermal switch activates (opens), remove the heat gun and observe the following (record values):
   (1) Thermometer shall indicate 150 ±15 °F.
   (2) M3 shall indicate 22 ±3 Vdc.
g. Allow power supply to cool down. As soon as thermal switch resets (closes), observe the following (record values):
   (1) Thermometer shall indicate 130 ±13 °F.
   (2) Oscilloscope shall indicate as in step d. above.

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TEST METHOD 4 - Continued.

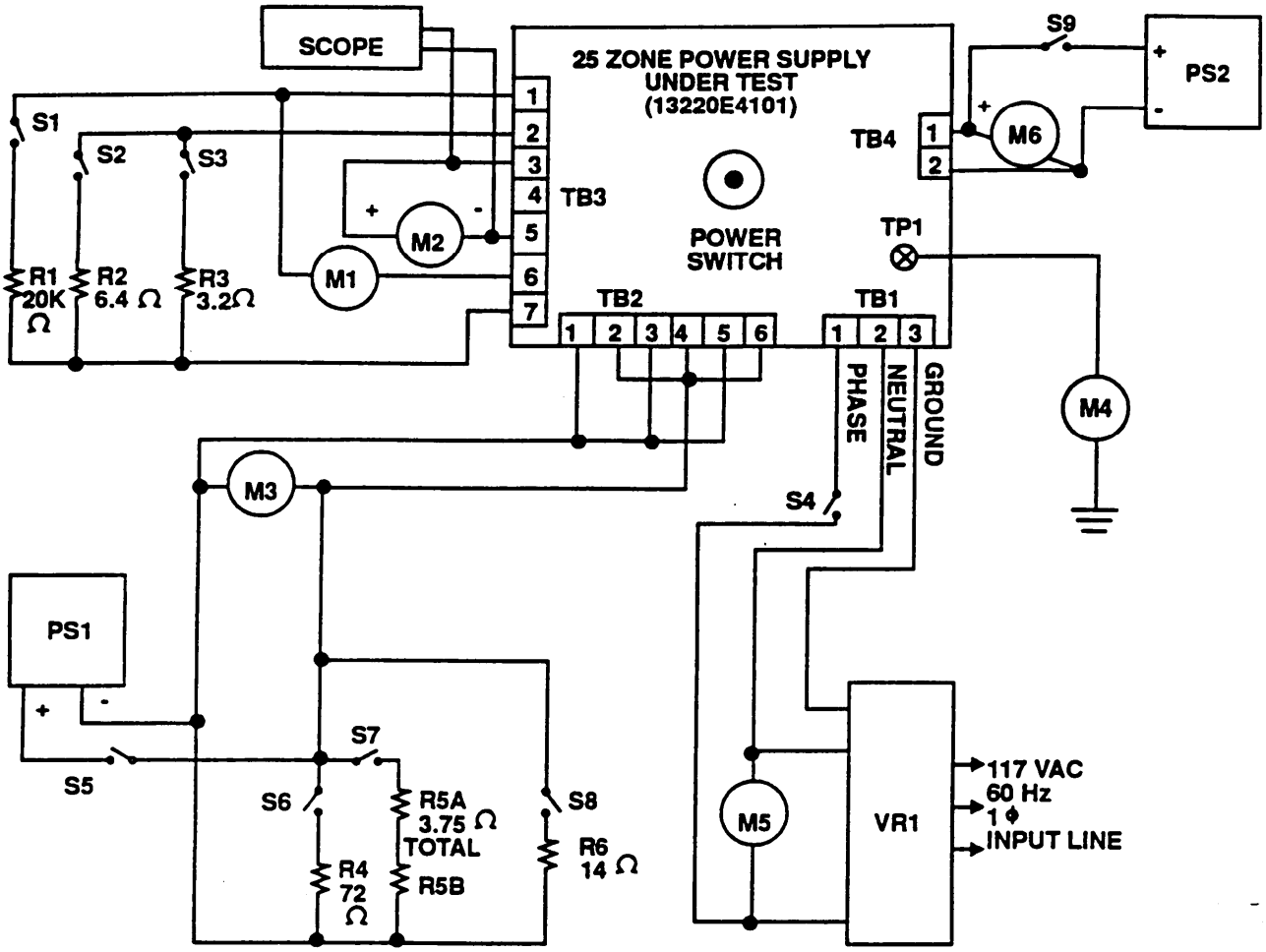
h. Replace power supply cover on PSUT.
i. For acceptance, power supply shall conform to above specifications.

3.7 Auxiliary dc power.

a. Open all switches on test fixture, except close S1, S3, and S9.
b. Adjust PS2 to 25 ±1 Vdc as indicated on M6.
c. M2 shall indicate 20 ±1 Vdc and M1 shall indicate 0 Vac.
d. For acceptance, measured values shall be within above specified limits.
e. If this power supply was removed from a monitor cabinet for testing, and no other functional tests are schedule for this unit, reinstall in monitor cabinet and verify proper connections by performing step 2.2 of test method 1.

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TEST METHOD 4 - Continued.



CALIBRATION PROCEDURE

To calibrate fixture, remove 25 Zone supply. Using a regulated power supply and an ammeter, calibrate each resistor in turn as follows (assume terminal strips labeled and numbered as shown, but without the power supply under test):

R3: With S3 closed, S2 open and an ammeter between the power supply and the lead going to TB3-2, set R3 for 20.0 Vdc on M2 with current of 6.25 A (+/- 3%).

R2: With S2 closed, S3 open and an ammeter between the power supply and the lead going to TB3-2, set R2 for 20.0 Vdc on M2 with current of 3.125 A (+/- 3%).

R4: With S6 closed, S5, S7 and S8 open and an ammeter between the power supply and the lead going to TB2-4, set R4 for 27.0 Vdc on M3 with current of 375 mA (+/- 3%).

R5: With S7 closed, S5, S6, and S8 open and an ammeter between the power supply and the lead going to TB2-4, set R5 for 30.0 Vdc on M3 with current 8.0 A (+/- 3%).

R6: With S8 closed and S5, S6 and S7 open and an ammeter between the power supply and the lead going to TB2-4, set R5 for 30.0 Vdc on M3 with current 2.14 A (+/- 3%).

R1: With S1 closed and an ammeter between the power supply and the lead going to TB3-1, select R1 for 22.5 Vac from TB3-1 to TB3-6 with current 1.125 mA (+/- 3%).

NOTES: a. Tolerances are 3% for current, 1% for voltage.
b. Negative lead of M4 shall be connected to any circuit ground.

FIGURE 3. Test fixture for 25 zone power supply.

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TEST METHOD 5

SIGNAL MODULE

1. TEST APPARATUS

a.	Digital voltmeter	3-1/2 digit or greater, minimum accuracy .25 percent of reading ± 1 LSD.
b.	Sound Level Meter	B & K type 2209 or equivalent.
c.	Test fixture	(See figure 4).

2. TEST PROCEDURE

2.1 General.

a. Close switch S1. Apply 20 Vdc to TB1-3. The signal module lamps shall be lit and the audible signaling device (ASD) shall be silent.
b. Open switch S1. After 3 seconds, the signal module lamps shall flash and the ASD shall sound.
c. Place the 'LAMP TEST/ACKNOWLEDGE' switch on signal module in the "ACKNOWLEDGE" position, the lamps shall extinguish, and the ASD shall be silent.
d. Place 'LAMP TEST/ACKNOWLEDGE' switch in the "LAMP TEST" position. The lamps shall light and remain lighted a long as switch is in "LAMP TEST" position. Release switch.
e. Close S1. The lamps shall flash and ASD shall sound. Place "LAMP TEST/ACKNOWLEDGE" switch in "ACKNOWLEDGE' position. The lamps shall stop flashing and remain on, and the ASD shall be silent.
f. Press switch S2. The ASD shall sound as long as switch is closed. Release S2. The ASD shall be silent.
g. Measure the sound pressure level of the ASD (weighting "A") at 10 feet directly in front of the ASD. The minimum acceptable sound pressure level shall be 60 dB.
h. If this signal module was removed from a monitor cabinet for testing, and no other functional tests are scheduled for this unit, reinstall in monitor cabinet and verify proper connections by performing step 2.2 of test method 1.

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TEST METHOD 5 - Continued

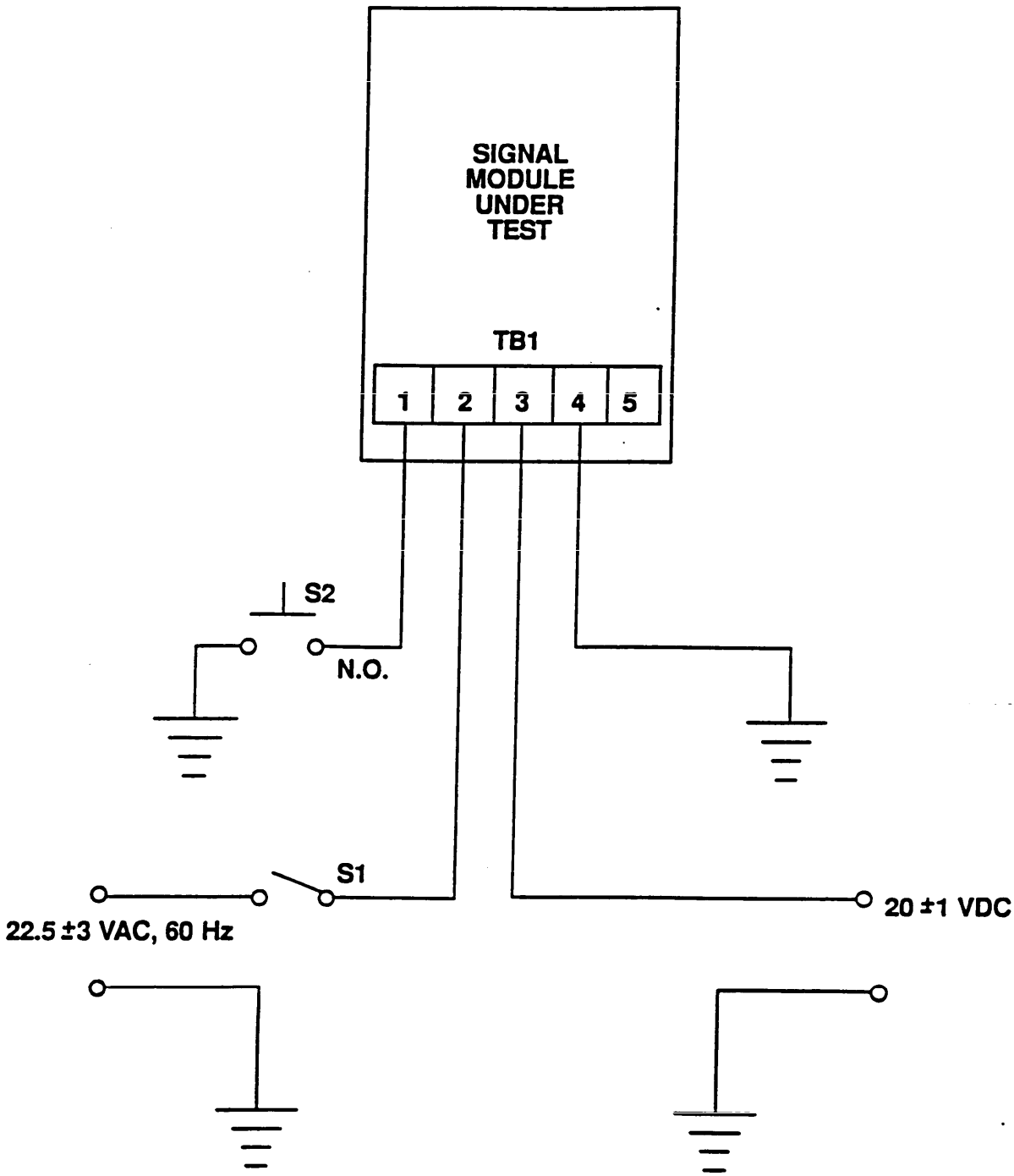


FIGURE 4. Test fixture for signal module

X-3888A

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APPENDIX B

POWER SUPPLY ALIGNMENT PROCEDURE

B.1 SCOPE

B.1.1 Scope. The procedure contained in this appendix is to assure proper power supply alignment. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

B.3 PROCEDURE

B.3.1 APPARATUS.

NOTE: See figure 1, 2, or 3 as applicable for apparatus and test figure configuration. Not all test instruments and components are required. However, the fixtures of figures 1, 2, and 3 may be used. One additional item not shown in the figures is a decade resistance box (or continuously variable resistance and an ohmmeter) adjustable over the range 19 K to 33 K ohms in increments not greater than 100 ohms.

B.3.2 ALIGNMENT PROCEDURE.

NOTE: This procedure shall be performed at room ambient temperature of 77 ±5 °F.

a. Connect decade resistor box in place of R41 on power supply PWB (A3 for 1 zone, A1 for 5 and 25 zone).
b. Set potentiometer R38 to mid-range and potentiometer R11 fully CCW.
c. Connect power supply under test (PSUT) and test equipment in accordance with figure 1, 2, or 3 as applicable. All test fixture switches and power switch of PSUT shall be open.
d. Close switch S4 and apply 117 ±3 Vac across TB1-1 (hot) and TB1-2 (neutral). Turn on power switch of PSUT. For 25 zone power supplies close S1 and S3 also. Allow at least 15 minutes for warm-up before proceeding.
e. Monitoring 27V output (TB2-2 to TB2-1) with DVM (M3), adjust R41 decade box to obtain following voltages:

1 zone P.S.: 27.9 ±0.2 Vdc
5 zone P.S.: 28.2 ±0.2 Vdc
25 zone P.S.: 27.7 ±0.2 Vdc

f. Monitoring 20 Vdc output (TB3-2 to TB3-5) with DVM (M2), adjust R38 to obtain 20.0 ±.1 Vdc.
g. Perform battery fast charge adjustment procedure as follows:

(1) With all test fixture switches off (open) and the PSUT switch turned off, turn on PS1 and adjust for 24 Vdc output (26 Vdc for 25 zone power supply).
(2) Turn R11 on power supply PWB of PSUT fully clockwise.
(3) Monitor TP1 on power supply PWB of PSUT with an oscilloscope or DVM.
(4) Close switch S5 on test fixture and turn PSUT power switch on. M3 shall indicate 24 Vdc (26 Vdc for 25 zone power supply). Adjust supply if necessary. Allow 5 minutes for warm-up stabilization.
(5) Observe voltage at TP1. It shall be 9 ±6 Vdc.
(6) Slowly adjust R11 CW until voltage at TP1 abruptly drops to less than 1 Vdc.
(7) Slowly increase PS1 output voltage until voltage at TP1 abruptly changes back to approximately the voltage observed in step e. the

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PS1 output voltage at the transition is the fast charge cut-out voltage. It should be 26.0 ±0.5 Vdc for the 1 and 5 zone supply and 28.0 ±0.6 Vdc for the 25 zone supplies. Record this voltage.

(8) Slowly decrease PS1 output voltage until voltage at TP1 abruptly drops to less than 1.0 Vdc. The PS1 voltage at the transition is the fast charge cut-in voltage. It shall be 24 ±0.2 Vdc for the 1 and 5 zone supplies and 26 ±0.2 Vdc for the 25 zone supply. Record this voltage.

(9) Repeat steps (7) and (8). Values noted should be the same as previously recorded, with a tolerance of 0.1 Vdc. If they are, record on acceptance test data sheet. If not, repeat until results are consistent.

NOTE: It may be necessary to repeat R11 adjustment if cut-in voltage appears out of tolerance.

(10) Successively check the 27V, 20V, and fast charge threshold voltage for appropriate values. If any are outside the tolerances specified above, repeat adjustments to the extent necessary.

(11) Select and install resistor R41, based on the decade resistance value determined in step (4) above.

NOTE: The requirement is that the power supply, when conformal coated and fully assembled, must meet the requirements specified in paragraph 3.1 of the appropriate test method of this appendix. Experience may show that R41 must have a slightly larger or smaller resistance than the value determined in step (5) above in order to meet the final functional test requirements after burn-in and conformal coating has been accomplished.

(12) Observe 27 Vdc output (M3). If voltage is not in accordance with step (5), re-select R41 until it is.

(13) Check 20 Vdc and fast charge threshold values. Adjust PSUT as necessary to bring within tolerance. If adjustments are required, re-check 27V output. Repeat iteration as necessary until all values are within tolerance.

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APPENDIX C

ENVIRONMENTAL TEST METHODS

C.1 SCOPE

C.1.1 Scope. The environmental test procedures contained in this appendix are to determine whether monitor cabinets and major components furnished separately procured under this specification conform to the environmental requirements specified. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

C.2 APPLICABLE DOCUMENTS

(This section is not applicable to the appendix.)

C.3 GENERAL REQUIREMENTS

C.3.1 Test conditions. Unless otherwise specified herein, tests shall be performed under the following standard ambient conditions:

Temperature: 77 ±5 °F (±15 °F when testing signal modules).
Relative humidity: 50 ±30 percent.

C.3.1.1 Measurements of test conditions. All measurements of test conditions shall be made with instruments of the accuracy specified in C.3.1.3.

C.3.1.2 Tolerance of test conditions. Unless otherwise specified, tolerance of test conditions shall be as follows:

a. Air temperature at the control sensor: ±2.5 °F. The equipment sensor response time (T) shall be 20 seconds or less. Temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5 °F per foot in any direction, but never more than 4 °F total (equipment nonoperating).

b. Relative humidity at the control sensor: ±5 percent.

NOTE: (T) is the time required for the sensing system to respond to 62.3 percent of a step change in temperature in the measured environment.

C.3.1.3 Accuracy of test apparatus. The accuracy of instruments and test equipment used to control or monitor the test parameters shall be verified and shall satisfy the requirements of ISO 10012-1 to the satisfaction of the procuring activity. All instruments and test equipment used in conducting the tests specified herein shall:

a. Conform to laboratory standards whose calibration is traceable to the prime standards at the US Bureau of Standards.

b. Have an accuracy of at least one-third the tolerance for the variable to be measured. In the event of conflict between this accuracy and a requirement for accuracy in any one of the test methods of this standard, the latter shall govern.

C.3.1.4 Stabilization of test temperature.

C.3.1.4.1 Operating. Unless otherwise specified, temperature stabilization will have been attained when the temperature of the part of the test item considered to have the longest thermal lag is changing no more than 3.6 °F per hour.

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C.3.1.4.2 Non-operating. Unless otherwise specified, temperature stabilization will have been attained when the temperature of the part of the test item considered to have the longest thermal lag reaches a temperature within 3.6 °F of the prescribed temperature. When changing temperatures, the temperature of the chamber air may be adjusted up to 9 °F beyond the desired end point for a period of time of up to 1 hour to reduce stabilization time, provided that the stabilization requirements of this paragraph are ultimately attained relative to the specified end point temperature, and provided the extended chamber temperatures will not cause damage to the test item.

C.3.2 Performance of test.

C.3.2.1 Functional test. The functional test referenced in these environmental test procedures shall be the same as that specified in the applicable subparagraph of 4.4.2.6, except as specified in the individual test procedures herein.

C.3.2.2 Pre-test functional test. Prior to proceeding with each environmental test, the test item shall be subjected to the applicable functional test (see C.3.2.1) under standard ambient conditions (see C.3.1). For the first environmental test (low temperature), this may be the functional test required after burn-in (see 4.4.2.4). For each subsequent environmental test (for example high temperature), this may be the post-test functional test after the previous environmental test.

C.3.2.3 Installation of test item in test chamber. The test chamber shall be at standard ambient conditions when the test item is installed. Suitable test leads shall be attached to each test item and brought out through port hole(s) to allow complete functional tests without opening the door of the test chamber during high temperature and low temperature tests. The test item shall then be subjected to the applicable functional test or an abbreviated functional test sufficient to determine that no malfunction or damage was caused due to faulty installation or handling. When testing monitor cabinets, power supply and signal module shall be removed from the monitor cabinet and each component tested in accordance with the applicable test procedures (testing during the same chamber run is optional).

C.3.2.4 Functional test at operating temperature. Perform applicable functional test (see C.3.2.1) during the test exposure as specified in the test procedure herein.

C.3.2.5 Post-test functional test. At the completion of each environmental test, the test item shall be subjected to the applicable functional test (see C.3.2.1) under standard ambient conditions (see C.3.1).

C.3.2.6 Failure criteria.

a. The item shall have failed the test when any of the following occur:

(1) Functional test results not in accordance with applicable test method during or after any of the tests.

(2) Deterioration, corrosion, damage or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational, service or maintenance requirements.

C.3.3 Test facilities and apparatus. Test facilities, chambers, and apparatus used in conducting the tests contained in this appendix shall be capable of meeting the conditions required.

C.3.3.1 Test chamber.

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C.3.3.1.1 Volume of test chamber. The volume of the test chamber shall be such that the bulk of the item under test will not interfere with the generation and maintenance of the test conditions. When testing multiple sample items simultaneously, the test chamber shall be of sufficient size so that each test unit is provided uniform environmental conditions and is not subjected to non-test environments.

C.3.3.1.2 Heat source. The heat source of the test facility shall be so located that radiant heat from the source will not fall directly on the test item.

C.3.3.1.3 Location of temperature sensors. Unless otherwise specified, thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be located centrally within the chamber, in the supply airstream, or in the return airstream, whichever provides the specified test conditions at the bulk under test and shall be baffled or otherwise protected against radiation effects.

C.3.3.1.4 Internal air circulation. The conditioned air flow shall be suitably baffled to provide uniform airflow around the test item. If multiple test items are tested, they shall be so spaced as to provide free circulation between the test items and the chamber walls.

C.3.4 Test data. Test data shall include complete identification of all test equipment and accessories. The data shall include the actual test sequence used and ambient test conditions recorded periodically during the test period. The test record shall contain a signature and date block for certification of the test data by the test engineer.

C.3.5 Test sequence. Units being subjected to environmental tests shall have been burned in, when applicable, and functionally tested at room ambient temperature prior to environmental tests. Environmental tests shall be performed in the sequence low temperature, high temperature, humidity and shock, unless otherwise approved by procuring contracting officer.

C.4 DETAILED TEST PROCEDURES

C.4.1 Low temperature test procedure.

C.4.1.1 Apparatus. Temperature chamber.

C.4.1.2 Procedure.

a. Prepare the test item(s) in accordance with General Requirements (see C.3).
b. Lower the internal chamber temperature to the storage temperature, -30 °F, and maintain for a period of 48 hours after stabilization.
c. Adjust the internal chamber temperature to -20 °F and maintain until temperature stabilization of the test item is reached or for one hour after the chamber temperature sensor has reached -20 °F.
d. Apply operating power to the test item until the item is stabilized (15 minutes).
e. Perform the functional test of Appendix A applicable to the test item.
f. Return the test item, nonoperating, to standard ambient conditions and maintain for 2 hours or until temperature stabilization of the test item is reached.
g. Apply operating power, allow 15 minutes for stabilization, and perform the functional test conducted in step e. above (post-low temperature test).

NOTE: The rate of temperature change during steps b., c., and f. may be the maximum attainable by the chamber but shall not exceed 18 °F per minute.

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C.4.2 High temperature test procedure.

C.4.2.1 Apparatus. Temperature chamber.

C.4.2.2 Procedure.

a. Prepare the test item in accordance with General Requirements (see C.3).
b. Raise the internal chamber temperature to 120 °F.
c. Maintain internal chamber temperature for 6 hours at 120 °F.
d. Raise the internal chamber temperature to 165 °F within a time period of 1 hour and then maintain at that temperature for 4 additional hours.
e. Lower the internal chamber temperature to 120 °F within a time period of 1 hour.
f. Repeat steps c., d., and e. two additional times (making a total of three 12-hour cycles).
g. Adjust the internal chamber temperature to +150 °F and maintain for 2 hours or until temperature stabilization of the test item is reached.
h. Apply power to the test item until the test item is stabilized (at least 15 minutes) and perform the functional test of Appendix A appropriate to the test item.
i. Return the test item, nonoperating, to standard ambient conditions and maintain for 2 hours or until temperature stabilization of the test item is reached.
j. Apply power to the test item until the item is stabilized (at least 15 minutes) and perform the test conducted in step h. above (post-high temperature test).

C.4.3 Humidity test procedure.

C.4.3.1 Apparatus. Humidity-temperature chamber and associated equipment.

C.4.3.1.1 Chamber. The chamber and accessories shall be constructed and arranged in such a manner as to avoid condensate dripping on the test item. The chamber shall be trap-vented to the atmosphere to prevent the buildup of total pressure. Relative humidity shall be determined from the dry bulb-wet bulb thermometer comparison method or an equivalent method approved by the contracting activity. When readout charts are used, they shall be capable of being read with a resolution within 1 °F. When the wet bulb control method is used, the wet bulb and tank shall be cleaned and a new wick installed at least every 30 days. The air velocity flowing across the wet bulb shall be not less than 900 feet per minute. Provisions shall be made for controlling the flow of air throughout the internal chamber test space where the velocity of air shall not exceed 150 feet per minute. Steam or distilled, demineralized, or de-ionized water having a pH value between 6.0 and 7.2 at 73 °F shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the test item by the test facility.

C.4.3.1.2 Procedure.

a. Prepare the test item in accordance with General Requirements (see C.3). Prior to starting the test, the internal chamber temperature shall be at standard ambient temperature (see C.3.1) with uncontrolled humidity.
b. Gradually raise internal chamber temperature to 149 °F and the relative humidity to 95 percent or greater over a period of 2 hours.
c. Maintain conditions of step b. for not less than 6 hours.
d. Maintain 92 percent, or greater, relative humidity and reduce internal chamber temperature in 16 hours to 86 °F. This temperature change shall be approximately linear with time and shall not exceed a change of 5 °F per hour. Seventy-five minutes before the end of the 16-hour period, commence performing a complete functional test of the

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test items in accordance with applicable test method of Appendix A. The test items may be removed from the environmental chamber for this test; however, unsuccessful completion of test or unsuccessful return of the test items to the test chamber within 75 minutes after initiation of tests shall constitute failure of the test. Failure to conform to the parameters specified in the functional test procedure shall also constitute failure of the test.

e. Repeat steps b., c. and d. for a total of 5 cycles (not less than 120 hours). Perform functional testing, as in step d. above, commencing 75 minutes before the completion of each 24-hour cycle. Figure 5 is an outline of the humidity cycle for this procedure.
f. After the completion of the fifth test period, return test items to standard ambient conditions, allow test item to reach temperature stabilization and perform functional test as in step d.

NOTE: The equipment shall be operated only when required tests are being performed.

C.4.4 Shock test (bench handling). This procedure shall be used to determine the ability of equipment to withstand the shock encountered during servicing. To test the monitor cabinet, remove the power supply and signal module and subject them to the test described herein. Remove the cover when testing the power supply. The test item shall be placed in a suitable position for servicing on a horizontal, solid wooden bench top at least 1-5/8 inches thick. The test shall be performed, as follows, in a manner simulating shocks likely to occur during servicing.

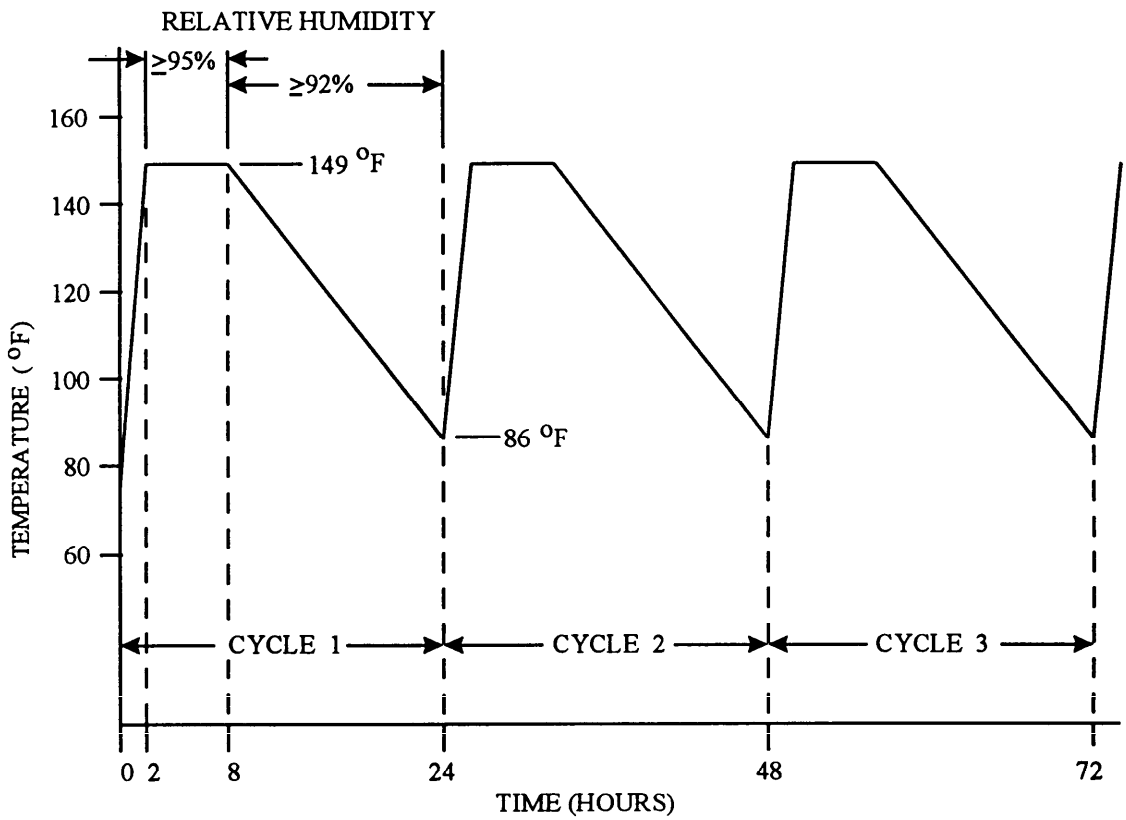
Step 1. Using one edge as a pivot, lift the opposite edge of the chassis until one of the following conditions occurs (whichever occurs first):

a. The chassis forms an angle of 45 degrees with the horizontal bench top.
b. The lifted edge of the chassis has been raised 4 inches above the horizontal bench top.
c. The lifted edge of the chassis is just below the point of perfect balance.

Let the chassis drop back freely to the horizontal bench top. Repeat, using other practical edges of the same horizontal face as pivot points, for a total of four drops.

Step 2. Repeat step 1, with the test item resting on other faces until it has been dropped for a total of four times on each face on which the test item could be placed practicably during servicing. The test item shall not be operating during the test. At the conclusion of the test, the test item shall be inspected and subjected to the applicable functional test of Appendix A. Failure criteria shall be as specified in C.3.2.5.

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NOTE: CONTINUE FOR A TOTAL OF 5 CYCLES (120 HOURS)

FIGURE 5. Test cycle for humidity test.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-52876D	2. DOCUMENT DATE (YYMMDD) 970630

3. DOCUMENT TITLE
Cabinets, Monitor, CY-7359/FSS-9(V) CY-7360/FSS-9(V), and CY-7361/FSS-9(V)

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)***

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